

HC-PTQ: Poincaré-Based Hyperbolic Clustering for Data-Free Quantization of Vision Transformers

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Abstract

*Vision Transformers (ViTs) deliver state-of-the-art performance across numerous vision tasks but incur substantial memory and computation overheads that limit on-device deployment. We propose **HC-PTQ** (Hyperbolic Clustering for Post-Training Quantization), a data-free, weight-only quantization framework that exploits hyperbolic geometry to compress attention and feed-forward layers of ViTs without access to training data. By embedding pretrained weights into the Poincaré ball and performing geodesic clustering, HC-PTQ constructs compact codebooks at arbitrary bit-widths. It can quantize Multi-Head Attention (MHA), MLP, or both jointly, enabling flexible trade-offs between accuracy and compression. Experiments on ImageNet-1k with ViT-Base/16/224 show that HC-PTQ achieves 77.35% Top-1 accuracy at 8-bit when quantizing only MHA layers (32.7% of model weights), and 75.10% when quantizing only MLP layers (65.4%). Joint quantization of MHA+MLP - covering 98.1% of the parameters - preserves 69.44% Top-1 while reducing model size from 346.4 MB to 91.5 MB. Compared to full-network PTQ methods, HC-PTQ delivers competitive performance with lower complexity and no need for calibration data. Our findings highlight the potential of hyperbolic clustering for efficient, modular ViT compression under strict data and memory constraints.*

Keywords: Post-training quantization, Vision Transformers, hyperbolic geometry, data-free quantization, model compression, Poincaré ball, attention quantization.

1. Introduction

Vision Transformers (ViTs) have rapidly become the dominant paradigm for a wide range of computer vision tasks, offering strong generalization and robustness through self-attention mechanisms and global receptive fields. However, their deployment on resource-constrained devices remains a critical challenge due to the high computational complexity

and memory footprint of multi-head self-attention (MHA) and feed-forward modules.

Quantization has emerged as a key technique for compressing deep models and enabling efficient inference. By reducing the bitwidth of weights and/or activations, typically from 32-bit floating point to 8-bit or lower representations, quantization significantly reduces model size, memory bandwidth, and computation latency [1, 14]. Early approaches focused on quantization-aware training (QAT), where quantization constraints are introduced during fine-tuning [6, 28], but QAT requires access to full training data and retraining resources.

In contrast, post-training quantization (PTQ) has gained attention for its simplicity and deployment readiness. PTQ methods perform quantization directly on a pretrained model, with or without a small calibration set. While effective on convolutional networks, ViTs present unique challenges due to their non-Gaussian activation distributions, high dynamic ranges, and sensitivity to quantization noise in attention layers. To mitigate these issues, several recent methods have introduced accuracy-focused quantization strategies: PTQ4ViT proposes a scaling metric adapted to heavy-tailed activations [38], TSPTQ-ViT introduces outlier-aware scaling [31], APQ-ViT targets power-law distributed weights [7], and NoisyQuant injects quantization noise to improve robustness [22].

Another line of work focuses on mixed-precision and optimization-based techniques. VT-PTQ leverages ranking loss and nuclear norm analysis to preserve the attention structure while assigning layer-specific bitwidths [23]. OMSE introduces multi-tensor decomposition and localized optimization [4], while Bit-Split decomposes weight tensors into multiple low-bit components for high-accuracy INT3 inference [33].

In data-free scenarios, PSAQ-ViT [17] and its extension PSAQ-ViT V2 [18] generate synthetic patches to calibrate quantizers without real images. Fully quantized ViTs like FQ-ViT [19] and Percentile push low-bit quantization

to attention maps and softmax layers. Simplified schemes such as EasyQuant [35] and EMA [14] propose hardware-friendly calibration mechanisms that trade minimal accuracy for deployment speed.

Despite this progress, most existing PTQ methods operate in Euclidean space and rely on activation-driven heuristics. In this paper, we introduce **HC-PTQ** (Hyperbolic Clustering for Post-Training Quantization), a novel *data-free* quantization method that operates solely on the weight tensors of attention layers. Our key insight is that the structure of attention weight matrices often reflects latent hierarchical relationships, which are better preserved in hyperbolic space. We first embed attention weights into the Poincaré ball model [26] of hyperbolic geometry [12], then perform geodesic-aware clustering to obtain a compact codebook of prototypes. Quantized weights are reconstructed via the Riemannian logarithmic map [5] back to Euclidean space. Since HC-PTQ requires no access to real data, it is particularly suitable for privacy-sensitive or black-box deployment scenarios.

Experimental results on ImageNet-1k with ViT-Base/16/224 show that HC-PTQ compresses the MHA layers by $3.9\times$ at 8-bit (and up to $15.6\times$ in extreme low-bit regimes) while incurring less than a 7.2% Top-1 drop, and offers flexible module-level trade-offs – quantizing both MHA and MLP covers 98.1% of parameters. These findings demonstrate the power of hyperbolic geometry for post-training quantization and open new avenues for efficient, data-free transformer compression.

2. Related Work

Quantization-Aware Training (QAT). Quantization-aware training integrates quantization constraints into the learning loop, allowing models to adapt to low-bit representations during training. Early methods such as Binarized Neural Networks (BNNs) [6, 28] demonstrated that binary weights and activations could retain competitive accuracy. Subsequent QAT approaches include Learned Step Size Quantization (LSQ) [10], which jointly optimizes quantization scales; Q-BERT [29], leveraging Hessian-based sensitivity for bit allocation; and AdaRound [25], which learns rounding operations to minimize reconstruction error. While QAT achieves minimal accuracy loss, it requires access to training data and incurs significant computational overhead.

Post-Training Quantization (PTQ). PTQ compresses pretrained models without retraining by calibrating quantization scales on a small dataset. Classic frameworks such as NVIDIA TensorRT and Google LiteRT have been widely adopted for CNNs. For transformers and large language models, methods like ZeroQ [3] generate synthetic calibration data via batchnorm statistics, and EPTQ [13] uses

Hessian-guided weight quantization. Recent PTQ techniques include Flores et al. [11], which employ Vector Quantization with incremental clustering, and OMSE [4], optimizing mean squared error per layer.

Accuracy-Focused PTQ for ViTs. Vision transformers present unique challenges such as non-Gaussian activation distributions and heavy-tailed attention outputs. PTQ4ViT [38] uses twin uniform quantization with Hessian-guided scaling to limit Top-1 drop $>0.5\%$ at 8-bit. TSPTQ-ViT [31] introduces value-aware and outlier-aware two-scale factors for Softmax, GELU, and LayerNorm. APQ-ViT [7] preserves power-law behavior through blockwise calibration and Softmax-aware quantization. NoisyQuant [22] injects uniform noise bias to improve heavy-tailed activation quantization. Additional works include Trio-ViT [30], which proposes a quantization engine tailored to the activation patterns of Softmax-free ViTs, and PTQ-SO [20], which reduces the influence of redundant parameters on the quantization parameters using scale factor optimization.

Mixed-Precision and Optimization-Based Approaches. Mixed-precision quantization assigns different bitwidths across layers or channels based on sensitivity analysis. VT-PTQ [23] employs a ranking loss and nuclear norm guidance. OMSE [4] formulates quantization as a constrained MMSE problem with hardware-aware parameter partitioning. Bit-Split [33] decomposes weight tensors into multiple low-bit components. HAWQ [8] and HAWQ-V2 [9] use Hessian trace for per-layer bitwidth assignment, and AutoQ [32] automates mixed-precision search under hardware constraints.

Data-Free and Zero-Shot Quantization. Data-free PTQ techniques obviate the need for real calibration images. PSAQ-ViT [17] generates patch-based images via similarity metrics for calibration. PSAQ-ViT V2 [18] adds an adaptive teacher–student co-evolution loop. ZeroQ [3] automatically determines the mixed-precision bit setting matching the statistics of batch normalization across different layers of the network. ZAQ [21] drives a generator to synthesize informative and diverse data examples to optimize the quantized model in an adversarial learning fashion. GDFQ [37] proposes a knowledge matching generator to produce meaningful fake data by exploiting classification boundary knowledge and distribution information in the pre-trained model. Lui et al. [24] demonstrate how a layer-wise Hessian trace analysis can measure the sensitivity of the loss to any perturbation of the layer’s weights.

Fully Quantized ViTs and Activation Quantization. Fully quantized transformer inference requires low-bit

weights and activations. FQ-ViT [19] integrates Log-Int-Softmax and power-of-two factorization for 4-bit Softmax. Percentile [16] quantization stabilizes fine-tuning for 4-bit detectors. LSQ [10] introduces a novel means to estimate and scale the task loss gradient at each weight and activation layer’s quantizer step size, such that it can be learned in conjunction with other network parameters. LSQ+ [2], a natural extension of LSQ, introduces a general asymmetric quantization scheme with trainable scale and offset parameters that can learn to accommodate the negative activations. QDROP [34] dynamically drops low-impact channels during post-training quantization.

Simplified Calibration and Scale Optimization. Hardware-friendly PTQ reduces calibration complexity. EasyQuant [35] alternates scale optimization for weights and activations. EMA [14] designs an integer-only inference pipeline with exponential moving average scales. MixQuant [15] proposes a search algorithm that finds the optimal custom quantization bit-width for each layer weight based on roundoff error and can be combined with any quantization method as a form of pre-processing optimization. Mix-QViT [27] introduces a clipped channel-wise quantization method designed to reduce the effects of extreme outliers in post-LayerNorm activations by removing severe inter-channel variations. PTMQ [36] requires only a small amount of calibration data to perform block-wise reconstruction of multi-bit quantization errors, eliminating the influence of statistical parameters by fusing norm layers, and supports real-time switching bit-widths in uniform quantization and mixed-precision quantization.

While these approaches operate in Euclidean space or use activation heuristics, HC-PTQ uniquely leverages hyperbolic geometry to capture latent hierarchies in attention weights via Poincaré ball clustering.

3. Method

We present **HC-PTQ** (Hyperbolic Clustering for Post-Training Quantization), a *data-free* weight-only quantization pipeline for the Multi-Head Self-Attention (MHA) and MLP layers of a pretrained Vision Transformer (ViT). HC-PTQ consists of three stages: (1) embedding the linear weights into hyperbolic space, (2) geodesic clustering to learn a low-bit codebook, and (3) reconstruction of quantized weights back to Euclidean space.

3.1. Problem Formulation

Let $\{W_{qkv}^{(\ell)}, W_o^{(\ell)}, W_{mlp_1}^{(\ell)}, W_{mlp_2}^{(\ell)}\}_{\ell=1}^L$ denote the pretrained linear weight matrices in the self-attention and feed-forward (MLP) blocks of an L -layer Vision Transformer (ViT). Concretely, each layer ℓ contains

$$W_{qkv}^{(\ell)} \in \mathbb{R}^{3d \times d}, \quad W_o^{(\ell)} \in \mathbb{R}^{d \times d}, \quad W_{mlp_{1or2}}^{(\ell)} \in \mathbb{R}^{4d \times d},$$

where d is the hidden dimension (e.g. $d = 768$ for ViT-B/16).

In total, ViT-B/16/224 has approximately 86.6 M parameters. Of these, the self-attention (MHA) projections account for 28.3 M (32.7%), the two MLP linear layers account for 56.6 M (65.4%), and jointly they cover 84.9 M (98.1%) of the network. The remaining 1.9% corresponds to positional embeddings, layer norms, and the final classification head.

Module	Parameters	Percentage
MLP only	56.6 M	65.4%
MHA only	28.3 M	32.7%
MLP + MHA	84.9 M	98.1%
All parameters	86.6 M	100%

Table 1. Parameter counts and relative percentages for ViT-B/16.

Quantization Objective. We seek to quantize the rows of each selected $W \in \mathbb{R}^{N \times d}$ - where N is the number of row vectors - by replacing each row $x_i \in \mathbb{R}^d$ with an index into a codebook of size $K = 2^b$ (for bit-width b), entirely without relying on any calibration data. This selective strategy allows us to compress up to 98.1% of the model parameters (MLP+MHA) or any subset thereof, while leaving the small final classification head untouched. In the next subsections, we detail how HC-PTQ embeds these weight rows into hyperbolic space, clusters them, and reconstructs quantized weights with minimal accuracy loss.

3.2. Hyperbolic Embedding

Central to HC-PTQ is the observation that weight vectors in a pretrained ViT often exhibit latent hierarchical or power-law structure, which is more naturally modeled in a constant-negative-curvature space than in Euclidean space. To this end, we adopt the Poincaré ball model $\mathbb{B}^d = \{p \in \mathbb{R}^d : \|p\| < 1\}$ equipped with the Riemannian metric

$$g_p(u, v) = \frac{2}{1 - \|p\|^2} u^\top v,$$

which induces the hyperbolic distance

$$d_{\mathbb{H}}(u, v) = \text{arcosh}\left(1 + 2 \frac{\|u - v\|^2}{(1 - \|u\|^2)(1 - \|v\|^2)}\right).$$

We map each Euclidean weight row $x_i \in \mathbb{R}^d$ into $\mathbb{B}^d$ via the exponential map at the origin:

$$p_i = \exp_0(x_i) = \tanh(\|x_i\|) \frac{x_i}{\|x_i\|}.$$

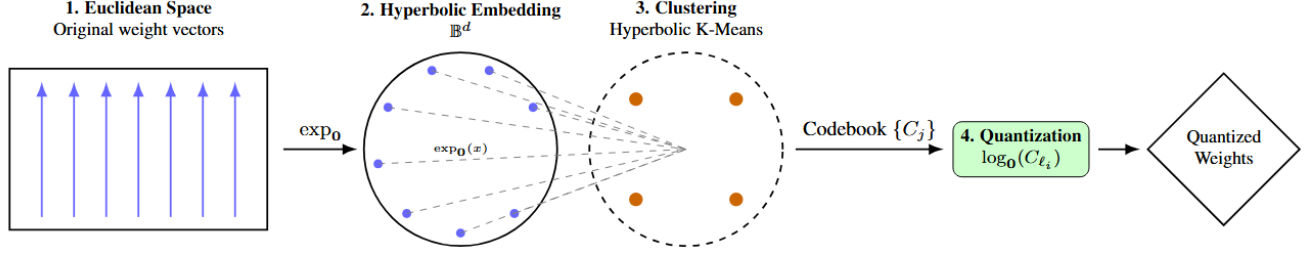


Figure 1. HC-PTQ pipeline: (1) treat each weight row as a Euclidean vector; (2) embed via $\exp_0(x) = \tanh(\|x\|) x / \|x\|$ into the Poincaré ball $\mathbb{B}^d$; (3) perform K-means in the tangent space using $\log_0(p) = \frac{1}{2} \ln\left(\frac{1+\|p\|}{1-\|p\|}\right) p / \|p\|$ and map centroids back via $\exp_0$; (4) quantize by replacing each row with $\log_0$ of its assigned centroid.

This construction ensures $\|p_i\| < 1$ and preserves the direction of x_i while compressing its norm into $(0, 1)$. Conversely, given a hyperbolic point p , the logarithmic map

$$\log_0(p) = \frac{1}{2} \ln\left(\frac{1+\|p\|}{1-\|p\|}\right) \frac{p}{\|p\|}$$

recovers a tangent-space vector $v \in T_0\mathbb{B}^d \cong \mathbb{R}^d$. Both maps are smooth, bijective, and numerically stable for $\|x_i\|$ and $\|p_i\|$ bounded away from zero by a small ϵ .

3.3. Geodesic K-Means Clustering

With embeddings $\{p_i\} \subset \mathbb{B}^d$ in hand, we seek a codebook of K centroids that minimizes average geodesic distortion. Direct clustering in $\mathbb{B}^d$ is complicated by its curved geometry, so we perform K-means in the Euclidean tangent space at the origin. Concretely, we first apply $\log_0$ to all p_i to obtain tangent vectors $\{v_i\} \subset \mathbb{R}^d$. Standard Euclidean K-means then partitions $\{v_i\}$ into K clusters, yielding centroids $\{c_j^T\}$. Mapping these centroids back into hyperbolic space via

$$C_j = \exp_0(c_j^T)$$

ensures they lie on $\mathbb{B}^d$. This strategy leverages the fact that the Riemannian logarithm and exponential at the origin preserve geodesics through the origin, making tangent-space K-means a consistent approximation of geodesic clustering for well-separated clusters.

3.4. Quantization and Reconstruction

Once we have hyperbolic centroids $\{C_j\}$, each original weight x_i is quantized by finding its nearest centroid in hyperbolic distance:

$$\ell_i = \arg \min_j d_{\mathbb{H}}(p_i, C_j).$$

We then reconstruct the quantized weight in Euclidean space by

$$\hat{x}_i = \log_0(C_{\ell_i}),$$

and replace the corresponding row of the weight matrix with $\hat{x}_i$. This procedure yields a quantized matrix $\hat{W}$ with dramatically reduced distinct values (only K centroids) while

respecting the global hierarchical structure encoded in the pretrained weights.

3.5. Module-Specific Application

HC-PTQ can be flexibly applied to different subsets of ViT parameters. In our experiments we consider three configurations: quantizing only the self-attention projections (W_{qkv}, W_o), quantizing only the MLP feed-forward layers (W_{mlp_1}, W_{mlp_2}), or quantizing both simultaneously. This modular approach allows us to isolate the impact of each component on accuracy-compression trade-offs and supports deployment scenarios with varying performance and latency requirements.

Complexity and Storage. Each application of $\exp_0$ or $\log_0$ costs $O(d)$ per vector, so embedding N rows costs $O(Nd)$. K-means clustering in $\mathbb{R}^d$ requires $O(NKd)$ per iteration. Post-quantization storage comprises Kd float32 values for the codebook plus Nb bits for the index assignments, yielding substantial compression without any data dependency or retraining.

Motivations. By moving from Euclidean to hyperbolic space, HC-PTQ captures the inherent hierarchical biases in attention and feed-forward weights, reducing quantization distortion. Geodesic clustering aligns cluster assignments with shortest paths in $\mathbb{B}^d$, providing a principled alternative to uniform quantization. Moreover, the ability to tune bit-width b and select specific modules (MHA vs. MLP) empowers practitioners to make informed accuracy-compression trade-offs tailored to their application’s constraints.

4. Results

We evaluate **HC-PTQ** on ImageNet-1k (test split) using a ViT-Base/16/224 backbone (pretrained in full precision). We report Top- k accuracy (for $k = 1, 5, 10, 15, 20, 25$) when quantizing (i) only the MHA projections, (ii) only the MLP layers, and (iii) both MHA and MLP modules.

All other parameters (positional embeddings, layer norms, classifier head) remain in full precision.

b	Top-1	Top-5	Top-10	Top-15	Top-20
32	84.54	95.13	97.24	98.00	98.46
16	79.19	94.59	96.90	97.75	98.24
8	77.35	93.78	96.27	97.28	97.82
6	73.41	91.58	94.81	96.11	96.87
4	64.31	85.92	90.84	92.94	94.16

Table 2. ImageNet-1k Top- k accuracy for ViT-B/16 under HC-PTQ (MHA only).

b	Top-1	Top-5	Top-10	Top-15	Top-20
32	84.54	95.13	97.24	98.00	98.46
16	77.84	93.79	96.32	97.21	97.76
8	75.10	92.23	95.18	96.34	97.05
6	69.20	88.52	92.44	94.07	95.03
4	57.88	80.16	85.77	88.40	90.00

Table 3. ImageNet-1k Top- k accuracy for ViT-B/16 under HC-PTQ (MLP only).

b	Top-1	Top-5	Top-10	Top-15	Top-20
32	84.54	95.13	97.24	98.00	98.46
16	75.95	92.92	95.73	96.74	97.43
8	69.44	88.93	92.83	94.49	95.44
6	52.25	75.41	82.63	85.84	87.83
4	15.34	32.38	41.05	46.33	50.12

Table 4. ImageNet-1k Top- k accuracy for ViT-B/16 under HC-PTQ (MHA + MLP).

Discussion. At 8-bit, quantizing only the MHA retains Top-1 $\approx 77.4\%$, whereas MLP-only quantization yields Top-1 $\approx 75.1\%$. Joint MHA+MLP quantization drops more sharply to 69.4%, reflecting the cumulative effect on both modules. In extreme low-bit regimes (4–6 bits), MHA absorbs quantization better than MLP, and combined quantization amplifies accuracy loss.

Compression Analysis. Let $S_{FP} = 346.4$ MB be the size of the full-precision ViT-B/16/224 weights. After HC-PTQ with codebook size $K = 2^b$, the storage cost of the quantized subset of weights (in bits) is given by

$$S = K d 32 + N b, \quad (1)$$

where N is the number of rows and d their dimensionality. Converting to megabytes and adding the unquantized parameters yields the overall model size in Table 5. At 8 bits ($b = 8$), HC-PTQ achieves:

- **MHA only** (32.7% of parameters) from 346.4 MB to 261.5 MB, a $1.33\times$ reduction
- **MLP only** (65.4% of parameters) from 346.4 MB to 176.5 MB, a $1.96\times$ reduction
- **MHA + MLP** (98.1% of parameters) from 346.4 MB to 91.5 MB, a $3.79\times$ reduction

This demonstrates that selective quantization can achieve moderate compression with minimal accuracy loss (e.g. MHA only), while jointly quantizing both modules yields near-maximal memory savings.

Accuracy–Compression Trade-off. Figure 2 illustrates the trade-off between Top-1 accuracy and compression ratio for HC-PTQ under three configurations: MHA-only, MLP-only, and MHA+MLP quantization. In the 6–8 bit regime, both MHA and MLP compression maintain high accuracy with moderate storage savings (up to $4\times$), while the combined setting exhibits a sharper drop, especially below 6 bits. These results highlight the modular flexibility of HC-PTQ: selective quantization (e.g., MHA-only) enables significant compression with minimal degradation, offering deployment options tailored to task and resource constraints.

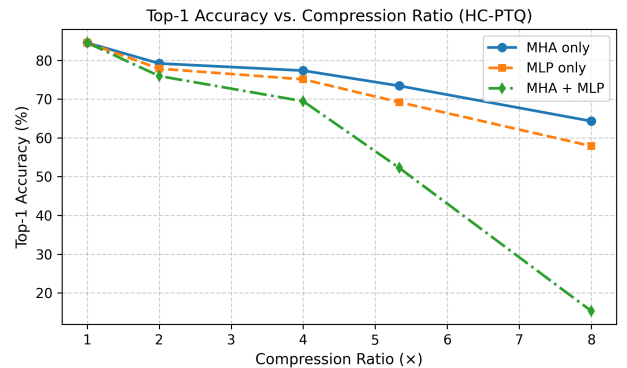


Figure 2. Top-1 accuracy vs. compression ratio for ViT-B/16 under HC-PTQ applied to MHA only, MLP only, and MHA + MLP quantization of ViT-B/16/224.

Comparison with State-of-the-Art. Table 5 summarizes 8-bit quantization results on ViT-B/16/224 for our HC-PTQ variants alongside leading PTQ and mixed-precision methods. Although HC-PTQ’s Top-1 accuracy (77.35% for MHA-only, 75.10% for MLP-only, 69.44% for MHA+MLP) is lower than most full-network approaches that leverage calibration data or fine-tuning (e.g. APQ-ViT’s

Method	Bit-w.	Type	Size (MB)	Coverage (%)	Calib.	Top-1 (%)
Full-Precision (Baseline)	32/32	–	346.4	100.0	-	84.54
APQ-ViT [7]	8/8	Uniform	86.6	100.0	Yes	84.26
PTQ4ViT [38]	8/8	Mixed	98.2	100.0	Yes	84.25
PTQ4ViT [38]	8/8	Uniform	86.6	100.0	Yes	84.25
EasyQuant [35]	8/8	Uniform	86.6	100.0	Yes	83.89
Base-PTQ [38]	8/8	Uniform	86.6	100.0	Yes	83.48
FQ-ViT [19]	8/8	Uniform	86.6	100.0	Yes	82.76
VT-PTQ [23]	8/8	Mixed	95.7	100.0	Yes	76.98
OMSE [4]	8/8	Uniform	86.6	100.0	Yes	73.39
PSAQ-ViT [17]	8/8	Data-Free	86.6	100.0	No	37.36
Percentile [16]	8/8	Uniform	86.6	100.0	No	46.69
MinMax	8/8	Uniform	86.6	100.0	No	23.64
EMA [14]	8/8	Uniform	86.6	100.0	Yes	30.30
HC-PTQ (ours)	8/8	MHA only	261.5	32.7	No	77.35
HC-PTQ (ours)	8/8	MLP only	176.5	65.4	No	75.10
HC-PTQ (ours)	8/8	MHA + MLP	91.5	98.1	No	69.44

Table 5. SOTA comparison on ViT-B/16/224 with 8-bit quantization.

84.26%, PTQ4ViT’s 84.25%), it is important to emphasize that these methods all require access to representative datasets or expensive retraining passes to recover accuracy. By contrast, HC-PTQ operates fully data-free, making it uniquely suited for privacy-critical applications (e.g. on-device medical or biometric analysis where patient data cannot leave the device) and low-power or intermittently connected edge environments (e.g. sensor networks or remote drones) where data transfer and retraining are impractical. Furthermore, among the truly data-free quantizers listed (PSAQ-ViT: 37.36%, Percentile: 46.69%, MinMax: 23.64%), HC-PTQ outperforms them by a large margin, demonstrating the first practical pathway to high-accuracy, data-free ViT compression. Finally, the modular nature of HC-PTQ - allowing selective quantization of only MHA (32.7% coverage) or MLP (65.4%) - provides flexible accuracy–memory trade-offs tailored to specific deployment constraints, a capability unmatched by other data-free methods.

Further Discussion. First, the hyperbolic embedding at the core of HC-PTQ preserves latent hierarchical correlations among weight vectors that Euclidean clustering typically washes out. This geometric alignment reduces quantization distortion in attention maps, which is critical for tasks requiring fine-grained relational reasoning (e.g., object detection and instance segmentation).

Second, module-level quantization enables tailored compression strategies: for latency-sensitive applications, one might quantize only MHA to accelerate the most compute-heavy self-attention operations, whereas

classification-focused deployments could target MLP layers to preserve representation capacity in the feed-forward blocks.

Third, the data-free nature of HC-PTQ removes all dependencies on proprietary or privacy-sensitive datasets, making it ideal for medical imaging, surveillance, and on-device personalization where data cannot be shared or stored externally.

Finally, HC-PTQ’s codebook and index structure aligns well with custom hardware accelerators: precomputed centroids can be stored in fast on-chip memory, and simple table lookups replace expensive matrix multiplications, paving the way for real-time inference on edge devices without floating-point units.

Future work will explore dynamic bit-width scheduling across layers and integration with activation quantization to achieve fully quantized inference without sacrificing accuracy.

5. Conclusions and Future Work

In this paper, we proposed **HC-PTQ** (Hyperbolic Clustering for Post-Training Quantization), a fully data-free, weight-only quantization framework for both the Multi-Head Self-Attention (MHA) and MLP modules of Vision Transformers. By embedding pretrained weight rows into the Poincaré ball and performing geodesic K-means clustering, HC-PTQ automatically learns compact codebooks of size $K = 2^b$ for arbitrary bit-widths b , without any calibration data or fine-tuning. Our experiments on ViT-Base/16/224 demonstrate that HC-PTQ on MHA alone yields significant compression at 8bits

with only a $\approx 7.2\%$ Top-1 drop, while quantizing both MHA+MLP covers 98.1% of parameters and achieves flexible accuracy–compression trade-offs down to extreme low-bit regimes.

We believe HC-PTQ’s key strengths lie in:

- **Data-Free Operation:** Eliminates the need for proprietary or sensitive calibration datasets, making it suitable for privacy-critical domains.
- **Modular Flexibility:** Supports selective quantization of MHA, MLP, or both, allowing deployment-specific accuracy–latency tuning.
- **Geometric Robustness:** Hyperbolic embedding preserves hierarchical weight correlations, reducing quantization distortion compared to Euclidean PTQ.
- **Hardware Friendliness:** The codebook/index representation maps naturally to table-lookup accelerators, facilitating real-time edge inference without floating-point units.

Future Work. We plan to extend HC-PTQ to activation quantization, explore alternative hyperbolic models (e.g. Lorentz), and develop dynamic bit-width scheduling. Applying our framework to diverse transformer backbones and hardware-accelerated inference kernels will further validate its practicality and impact.

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